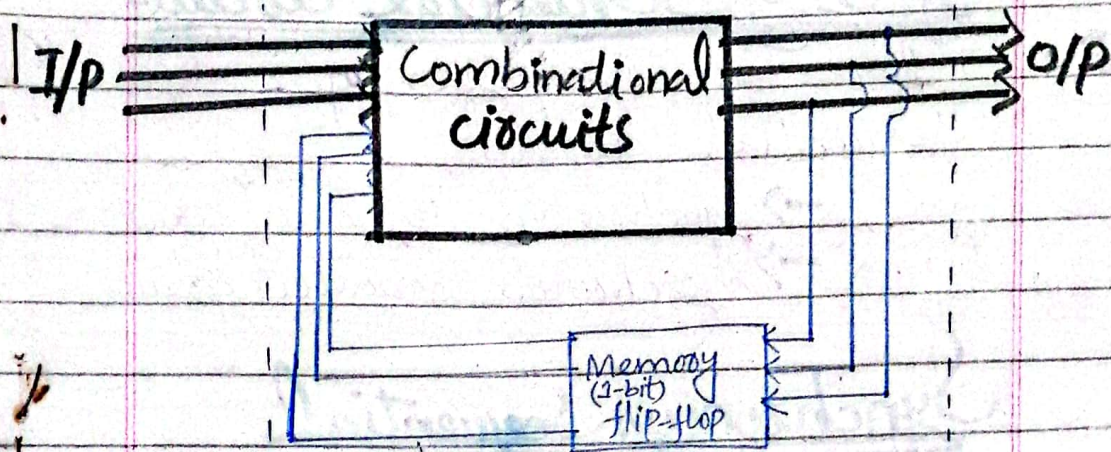


(Ch #7)

Sequential Circuits

→ Sequential circuits are combinational circuits with memory.

→ In Sequential circuits, the present o/p depends on the present I/P as well as previous output to produce output.



Sequential circuits

Give examples of
Sequential &
Combinational circuits

Combinational Circuits

- Adders
- Subtractors
- Multiplexers
- Encoders
- Decoders

Sequential Circuits

- Flip-flops
- Counters
- Registers

Types of Sequential Circuits

There are two types of sequential circuits:

- Synchronous sequential circuit
- Asynchronous sequential circuit

Synchronous Sequential Circuit

A sequential circuit whose output behaviour depends on the input at a discrete-time is called synchronous sequential circuit.

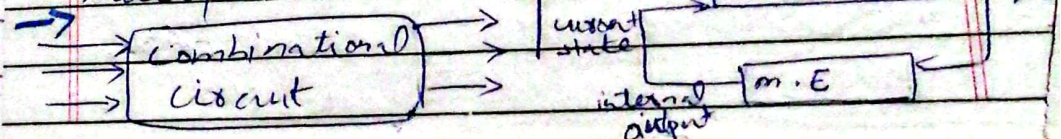
Asynchronous Sequential Circuit:

whose output depends on the sequence in which the input changes.

Date

What is difference between Combinational and Sequential circuits?

- | | |
|---|--|
| → Time independent | → Time dependent |
| → depend on the present input to generate output. | → depend upon present output as well as previous input |
| → Speed is fast | → Speed is slow |
| → Designed easily | → Designed tough |
| Used for arithmetic as well as boolean operation. | Used for elementary building blocks like flip-flop. |
| → Don't have clock | → Have clock |
| → Does not require triggering | → Require triggering |
| → No capability to store any state | → Have capability to store state |
| → Don't have memory element | → Have memory element |
| → Encoder, decoder, multiplexers, DEMUX | → Flip-flops, counters |



What is difference b/w Synchronous and Asynchronous circuit?

Synchronous Circuit

→ It is easy to design.

→ A clocked flip-flop acts as memory element.

→ They are slower as clock is involved.

→ The states of memory is updated only at active edge of clock, if input is changed.

→ These are more expensive.

→ It has complex circuitry.

→ Also called clocked sequential circuit.

Asynchronous Circuit

→ It is difficult to design.

→ An unlocked flip-flop or time delay is used as memory element.

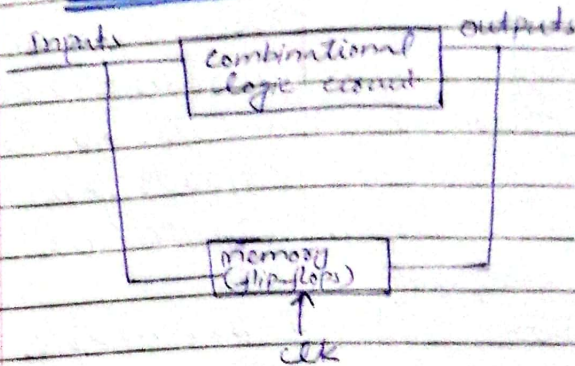
→ They are comprised of series or no clock is used here.

→ The states of memory element will change any time as soon as input is changed.

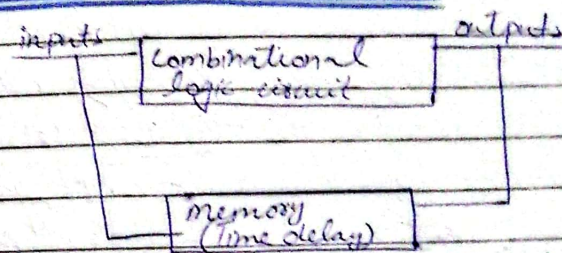
→ These are economical.

→ Also called event driven circuit.

Block diagram of Synchronous Sequential Circuit:-



Block diagram of Asynchronous Sequential circuit:-



What is race condition in Asynchronous circuit?

→ There are chances for the asynchronous circuits to enter into a wrong state because of the time differences b/w the arrival of inputs. This is called as race condition.

Latches

Latch:- Holds 0 or 1 Position.

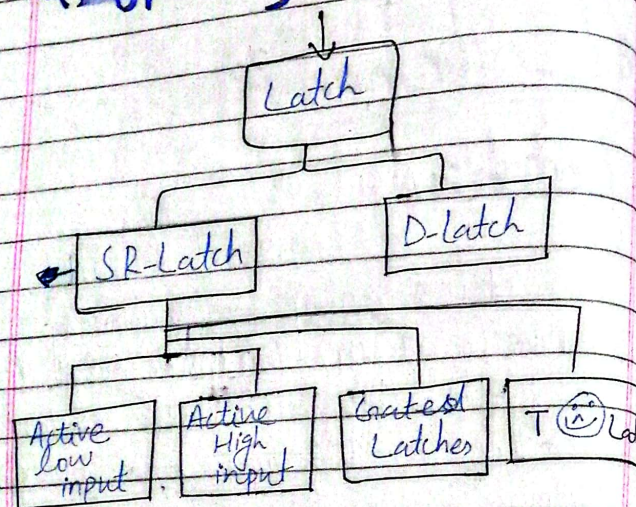
- Latch is Sequential circuit.
- Latch is a type of temporary storage device. Basic storage element.
- Latches have two stable states.
- There are two inputs and two outputs of the latch.
- The outputs of the circuit are complement of each other.
 $Q \neq \bar{Q}$

Definition:-

- Asynchronous circuit

- A cross coupled sequential circuit having bistable (set and reset) states; in which the output of the gate (1) is the input of the gate (2) and the output of the gate (2) is the input of the gate (1).

Types of Latches



SR Latches:-

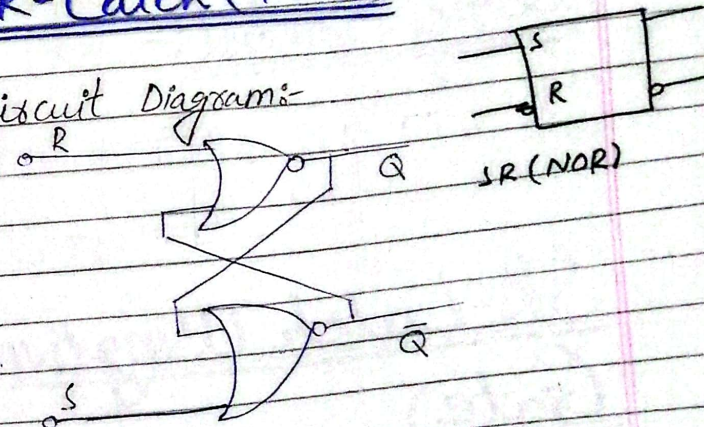
- The basic storage element. There are two inputs: R (Reset) and S (Set).
- There are two outputs Q and $\bar{Q}$. Where Q is never equal to $\bar{Q}$ and both are complement of each other.
- The latch of NOR gate is also called Active high input latch.

→ The latch of NAND gate is also called Active low input latch.

- if Q=1 the condition is set
- if Q=0 the condition is reset.

SR-Latch (NOR):-

Circuit Diagram:-



Truth table for NOR gate:-

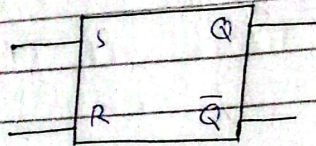
A	B	A+B output	A+B
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

Case I:- $S=0, R=1, Q=0, \bar{Q}=1$
 $S=0, R=0, Q=0, \bar{Q}=1$

Case II:-
 $S=1, R=0, Q=1, \bar{Q}=0$
 $S=0, R=1, Q=1, \bar{Q}=0$

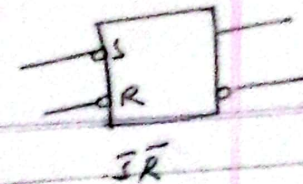
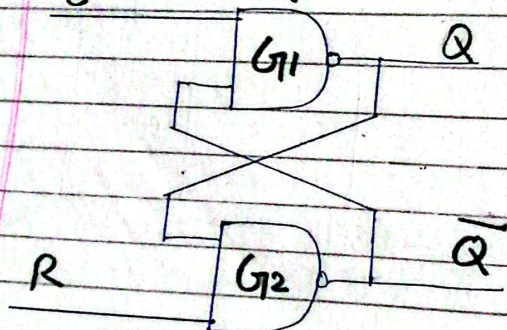
Case III:-
 $S=1, R=1, Q=0, \bar{Q}=0$
 $R=0, Q=0, Q=0, \bar{Q}=1$
 Q & $\bar{Q}$ should be complemented of each other.

Logic Symbol:-



SR-Latch Using (NAND Gate)

Circuit Diagram:-



Truth table:-

NAND gate T.T

A	B	$X=A \cdot B$	$\overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

$Q = (n+1)$
 NOT
 State

Case I:-

$S=0, R=0, Q=1, \bar{Q}=1$
 $S=1, R=1, Q=1, \bar{Q}=0$
 Not used
 $S=1, R=1, Q=0, \bar{Q}=1$

Case II:-

$S=1, R=0, Q=0, \bar{Q}=1$
 $S=1, R=1, Q=0, \bar{Q}=1$

Case III:-

$S=0, R=1, Q=1, \bar{Q}=0$
 $S=1, R=1, Q=1, \bar{Q}=0$ (Active low)

Case IV:-

$S=1, R=1, Q=1, \bar{Q}=0$

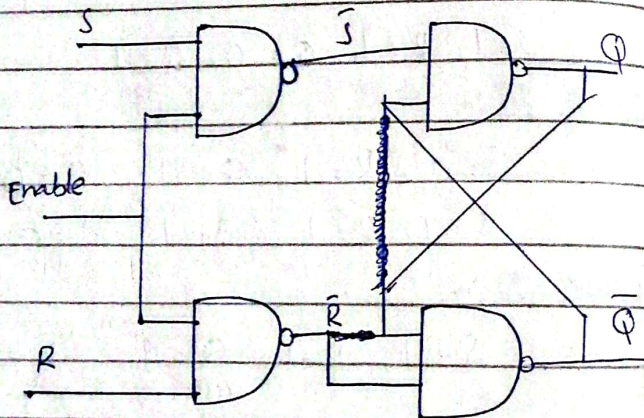
Gated S-R Latch

- working is similar to SR latch.
 → A gated latch requires an enable input, En (G) is also used to designate an enable input.

→ The gated latch is a level-sensitive device.

→ In this circuit, the invalid state occurs when both S and R simultaneously High and EN is also High.

Circuit Diagram



→ The enable determines when the state of the latch can be change.

S	R	E	Q	$\bar{Q}$
0	0	1	Q	$\bar{Q}$
0	1	1	0	1
1	0	1	1	0
1	1	1	(ignore)	
X	X	0	Q	$\bar{Q}$

∴ $Q \neq \bar{Q}$

D-Latch (Transparent Latch)

→ D latch eliminates the undesirable condition of the intermediate state that occurs in the SR latch.

→ The D latch stands for data latch.

→ The D latch has one input but two outputs.

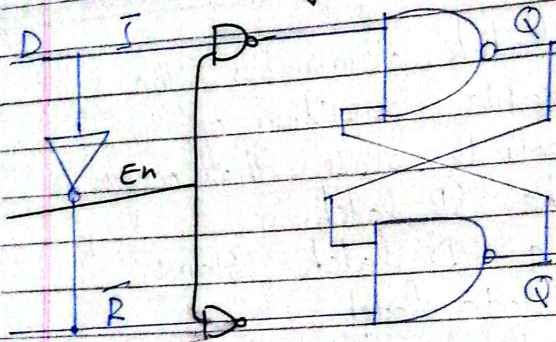
→ NOT gate is used in the latch.

→ D latch removes the invalidity of the SR latch.

→ Means D-latch gives correct complement inputs on the $S=1$ and $R=1$ in NOR Gate latch.

→ And $S=0$ and $R=0$ in NAND gate latch.
 → if $Q=1$ (high) then the latch is the set state.
 → if $Q=0$ then the latch is the Reset state.

Circuit Diagram



Truth table:-

D	Q	$\bar{Q}$
0	0	1
1	1	0

O/p = i/p = $En=1$

O/p = mm, $En=0$

What race around condition in JK flip-flop.

→ Change of output without change in input.

→ When $J=1$ and $K=1$ then

race condition occurs.

What is propagation delay?

Time for appear output after input is applied.

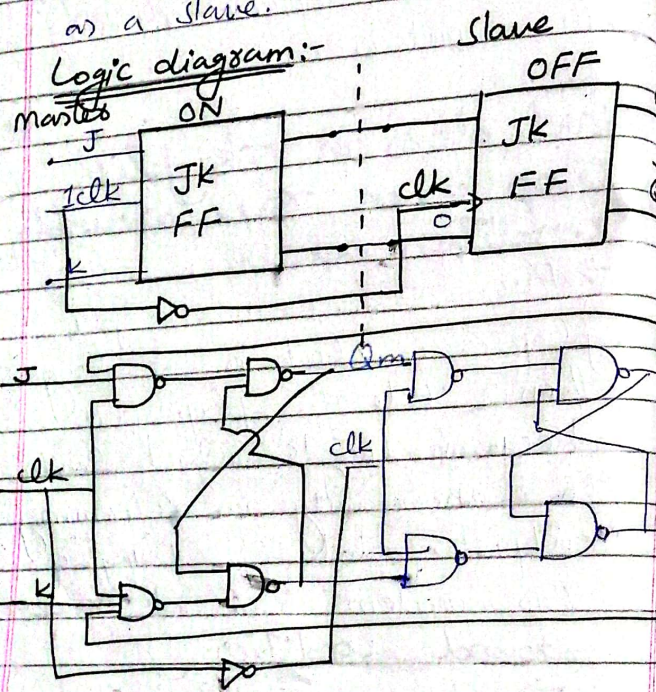
What does in JK flip flop when race around condition occurs?

→ For $J=K=1$ in JK flip flop & if clock is too long then state of flip flop keep on toggle which leads to uncertainty in determining output state of flip flop. This problem is called "race around condition".

→ To solve this problem we use Master Slave JK flip flop.

Draw Circuit Diagram of Master-Slave JK flip flop.

→ In master slave JK flip flop we use two JK flip flop.
 → 1st JK flip flop works as master and other flip flop works as a slave.



$J = K = 1, \text{Clk} = 1, Q_m = \text{Active}, Q_s = 1$
 $J = K = 1, \text{clk} = 0, Q_m = \text{memory}, Q_s = 1$

Why latches are called memory devices?

Latches can be memory devices, and can store one bit of data for as long as the device is powered. It is formed by interconnection of logic gates.

Why flip flops are preferred over latches?

Generally designers prefer flip flops over latches because of edge-triggered property, which makes the behaviour of the timing simple & ease design interpretation.

Which IC used for SR flip flop?

The IC used is SN74HC00N. It is a 14 pin package which contains 4 individual NAND gates.

What are the two models of/in synchronous circuits?

Mooré circuit

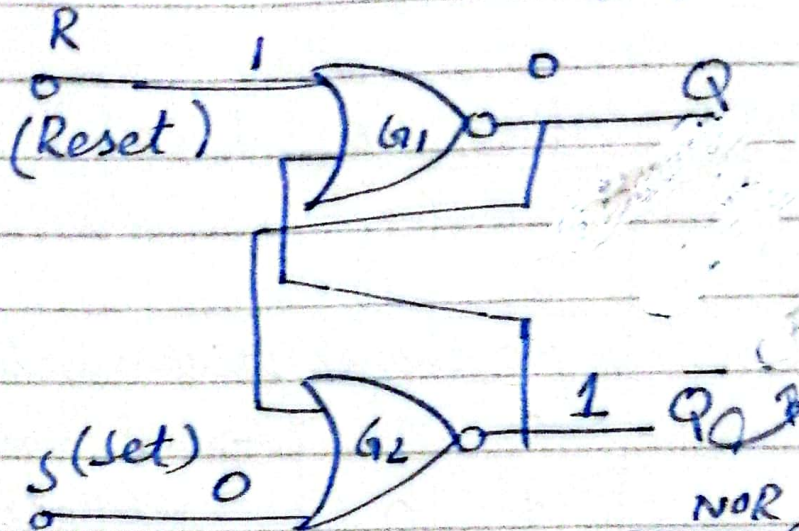
Maly circuit.

Latches are used as temporary buffers.
Latches $\rightarrow$ Basic storage element which holds '0' or '1' position.

SR Latch

$\rightarrow$ Asynchronous circuit

Using NOR gates:- behave as memory - if in any manner the input is change the output will also be change



Reset $Q = 0$
 Set $Q = 1$

Note: Any input is 1 the output will be zero.

NOR gate T.T

A	B	A+B
0	0	1
0	1	0
1	0	0
1	1	0

Case 1:-

$$S = 0, R = 1, Q = 0, \bar{Q} = 1$$

(~~Case 1~~)

$$S = 0, R = 0, Q = 0, \bar{Q} = 1$$

Case II

$$S = 1, R = 0, Q = 1, \bar{Q} = 0$$

$$S = 0, R = 0, Q = 1, \bar{Q} = 0$$

Case 1 $S=1, R=1, Q=0, \bar{Q}=0$
 Not true

$S=0, R=0 \begin{cases} Q=0, \bar{Q}=1 \\ Q=1, \bar{Q}=0 \end{cases}$

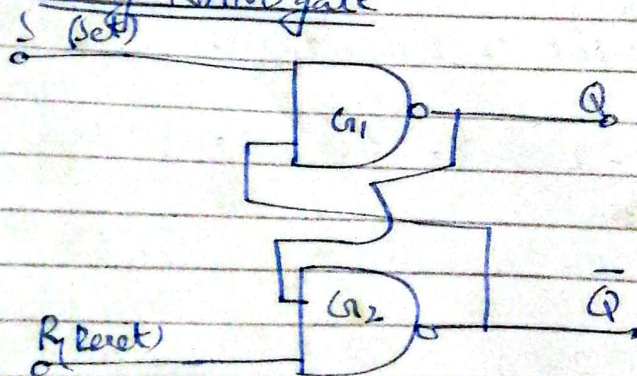
Contradictory state

$S=1, R=1 \rightarrow$ So we don't use them.

Truth table for S-R latch

S	R	Q	$\bar{Q}$
0	0	No change / previous state	
0	1	0	1
1	0	1	0
1	1	Not used	

Using NAND gate



any input is zero then output will be high

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Case I :-

$S=0, R=0, Q=1, \bar{Q}=1$
 $\rightarrow$ Not used state

Case II

$S=1, R=0, Q=0, \bar{Q}=1$

Case III :-

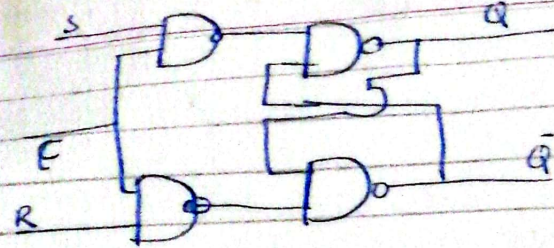
$S=0, R=1, Q=1, \bar{Q}=0$

Case IV :-

$S=1, R=1, Q = \text{prev}, \bar{Q} = \text{prev state}$
 $\rightarrow$ No change / previous state

S	R	Q	$\bar{Q}$
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Previous state	

Enabled SR latch



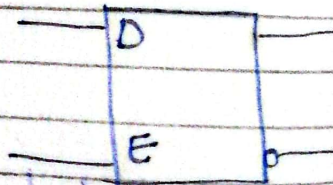
→ Its working is similar as SR latch with using NOR gate.

E	S	R	Next state Q
0	x	x	No change
1	0	0	No change
1	0	1	$Q=0$; Reset state
1	1	0	$Q=1$; Set state
1	1	1	Indeterminate

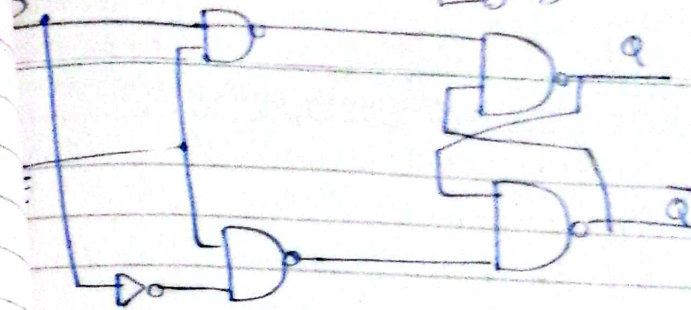
- The Enable E determines when the state of the latch can be changed.

Gate 1

D Latch:-



can change



∴ D latch eliminates the undesirable condition of the intermediate state that occurs in the SR latch ($Q = \bar{Q} = 1$)

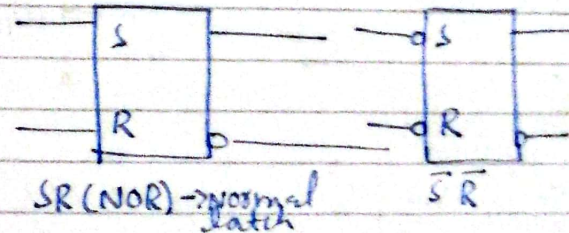
if $D=1, Q=1 \rightarrow$ Set state

if $D=0, Q=0 \rightarrow$ Reset state

(T.T table):-

E	D	Next state of Q
0	x	No change
1	0	$Q=0$, Reset state
1	1	$Q=1$, Set state

Symbols for latches



SR (NOR) $\rightarrow$ normal latch

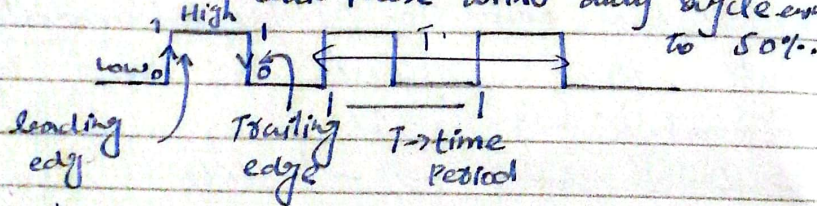
$\bar{S} \bar{R}$

Clock & Triggering Methods

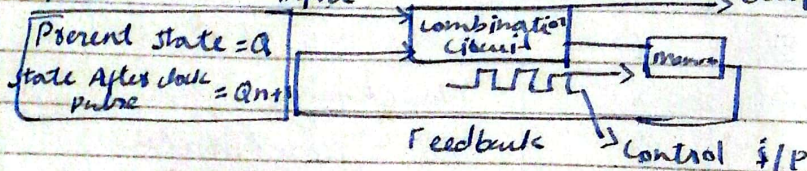
- clock is a signal that continuously oscillates b/w certain fixed level of V_H & V_L

Clock signal:- clock signal is a time signal. Also called clock pulse
- used to provide sequence of the ckt

- clock is a rectangular signal called clock pulse with duty cycle comes to 50%.

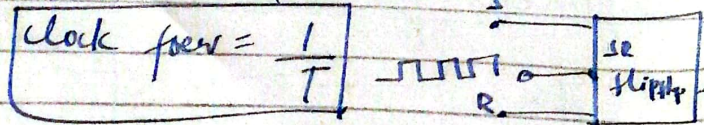


When we take input called transparent mode when we follow input o/p called latch mode



- clock signal work as control signal for sequential circuit

- is used to provide synchronization b/w i/p & o/p



Duty cycle:-

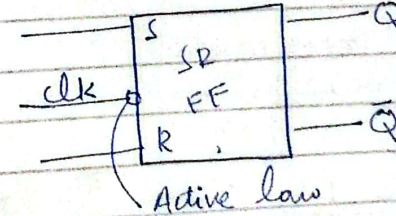
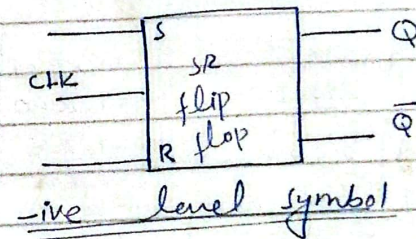
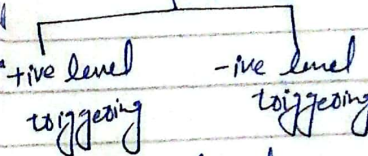
= ratio of time for which signal is high to total time

$$= \frac{t_H}{T} = \frac{1}{2} = 50\%$$

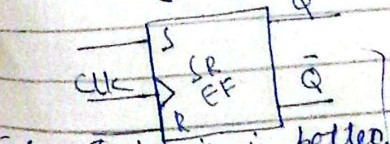


Triggering method

when a flip flop changes its state at least one time



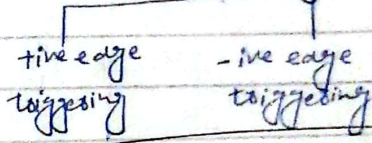
Active low



Edge Triggering is better than level triggering

A flip flop is said to be triggered when a trigger pulse is applied to the input that bring the change in output.

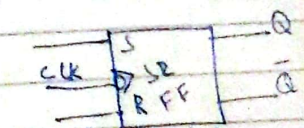
Edge Triggering



The change in output of a flip flop can be done by bringing a small change in the input signal. This small change can be done with help of clock pulse. This clock pulse is called trigger pulse.



-ive edge triggering



than level triggering